

ResQuick AI — Project Technical Report

Smart Disaster Response, GIS Ingress Routing & Mathematical Resource Prioritization

System: DisasterIQ Municipal Core	Date: October 2026	Document Type: Official Hackathon / Competition Technical Specification
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1. Executive Summary & Problem Statement

During severe urban weather emergencies, cloudbursts, and flash floods, civic control rooms are overwhelmed by thousands of disconnected emergency calls. Operators face severe bottlenecks: they lack real-time visibility into which roads are submerged, struggle to communicate with citizens who do not speak English, and have no objective mathematical methodology to determine which trapped citizens or critical hospitals require immediate tactical resources.

ResQuick AI resolves this life-or-death crisis by providing a dual-portal ecosystem (Citizen Assistance Portal + Municipal Commander Command Center). It fuses **browser-native multi-language speech recognition, automated NLP semantic translation to English, high-performance GIS street mapping with alternative ingress corridors**, and a **5-factor dynamic mathematical risk formula** that auto-ranks disaster zones and dynamically redistributes emergency resources in real-time.

2. Complete Technology Stack Used

Domain / Component	Technology & Framework	Purpose & Implementation Details
Frontend Core	React 18 with Vite 5	Component-driven architecture, ultra-fast Hot Module Replacement (HMR), optimized production asset chunking.
Styling & Design System	Tailwind CSS 3 & Glassmorphism	Adaptive light & tactical dark themes, custom backdrop blurs (backdrop-blur-md), mobile-responsive grid layouts.
Interactive GIS Map	Leaflet.js (v1.9)	Open-source geospatial mapping with synchronized viewport bounding, zero-lag tile invalidation, and custom DivIcons.
Street Map CDN	CartoDB Voyager (Cloudflare CDN)	High-speed, sub-40ms vector-rendered street map tiles showing exact Bengaluru roads, flyovers, metro corridors, and landmarks.
Satellite Layer	Esri ArcGIS World Imagery	High-resolution satellite raster imagery for terrain analysis and waterbody inspection.
Geolocation & GPS	HTML5 Geolocation + OSM Nominatim	Real-time browser coordinate acquisition with reverse-geocoding into clean street

Domain / Component	Technology & Framework	Purpose & Implementation Details
		addresses.
Voice Recognition (Mic)	HTML5 Web Speech Recognition API	Captures citizen speech directly via microphone in regional Indian locales (kn-IN, hi-IN, ta-IN, te-IN, etc.).
NLP Translation Engine	Bi-directional Multilingual NLP Pipeline	Tokenizes native regional speech transcripts and translates them in real-time into standardized English for government officials.
Audio & Sound System	Web Audio API / HTML5 Audio	Frequency-synthesized alert pings, emergency siren loops, and broadcast dispatch ringtones.
Iconography & UI	Lucide React	Lightweight, scalable vector icons for hazards, navigation, sensors, and status badges.

3. Languages Supported

ResQuick AI bridges the communication divide between regional citizens and administrative commanders across **8 official Indian languages + English**:

- **Kannada (ಕನ್ನಡ)** — [kn-IN]: Primary municipal language for Bengaluru civic grievances.
- **Hindi (हिन्दी)** — [hi-IN]: National language support for inter-state residents and NDRF squads.
- **Tamil (தமிழ்)** — [ta-IN]: Full regional speech and text translation.
- **Telugu (తెలుగు)** — [te-IN]: Full regional speech and text translation.
- **Malayalam (മലയാളം)** — [ml-IN]: Regional dialect support for southern transit corridors.
- **Marathi (मराठी)** — [mr-IN]: Supported for western regional citizens.
- **Bengali (বাংলা)** — [bn-IN]: Supported for eastern regional citizens.
- **English (English)** — [en-IN / en-US]: Universal administrative lingua franca for commanders.

4. Mathematical Risk Prioritization Model

Rather than relying on arbitrary operator decisions, ResQuick AI implements a formal **Multi-Criteria Decision Analysis (MCDA)** risk scoring engine. Every sector is continuously evaluated against five objective weighted parameters:

$$\text{Risk Score} = (0.30 \times S) + (0.25 \times P) + (0.20 \times I) + (0.15 \times A) + (0.10 \times C)$$

Variable	Parameter Name	Weight	Metric & Evaluation Criteria
S	Hazard Severity	30%	Physical depth of standing water (0–2m), building structural failure rate, fire flame velocity.
P	Population Exposure	25%	Count and demographic density of civilians, elderly, and patients trapped within the hazard polygon.
I	Infrastructure Damage	20%	Substation immersion, metro corridor shutdown, telecommunication tower collapse risk.
A	Accessibility Difficulty	15%	Degree of physical ingress blockage, culvert breaches, fallen trees, and debris obstruction.
C	Critical Facility Threat	10%	Proximity to lifeline facilities (hospitals, trauma units, oxygen supplies, drinking water plants).

Priority Score Tiers & Automated Action Rules

- Score 80 – 100: CRITICAL** — Immediate NDRF Tactical Water Squad dispatch, trauma ambulance rerouting, automated Siren Alert.
- Score 60 – 79: HIGH** — Fire & Rescue units deployed, heavy earthmovers tasked, alternative ingress routes opened.
- Score 40 – 59: MODERATE** — Municipal engineering dewatering pumps deployed, traffic diversions active.
- Score < 40: LOW / WATCH** — Continuous surveillance, preventative drain clearance, baseline watch.

5. Datasets & Schemas Used

Dataset Name	Storage File	Contents & Schema Attributes
Bengaluru Municipal Hazard Zones	src/data/zonesData.js	5 real-world wards (Indiranagar Halasuru, Bellandur Yamalur, Hebbal Flyover, Shivajinagar Russell Market, Peenya Phase 4). Attributes: coordinates, baseline risk scores, road blockage counts,

Dataset Name	Storage File	Contents & Schema Attributes
		vulnerable population numbers, critical facilities list, verified ground photo evidence.
Citizen Grievance Incident Registry	src/data/sampleIncidents.js	Live incident schema: Application ID (CS-2026-XXXXX), reporter info, category, original native speech transcript, translated English text, GPS lat/lng, assigned municipal department, photo forensics status, real-time lifecycle status (Submitted, Verified, In Progress, Resolved).
Smart Emergency Ingress Corridors	src/utils/resourceEngine.js	Graph pairing blocked primary arteries (with specific obstruction cause) against verified clear AI alternative routes with calculated distance and ETA.
Doppler Weather & Predictive Telemetry	WeatherForecastWidget.jsx	Simulated IMD radar telemetry: Temperature (24°C), rainfall rate (42 mm/hr), humidity (94%), wind velocity (38 km/h), and predictive 3-hour inundation forecasts.
National Emergency Helplines & Bulletins	src/data/indiaNews.js	Official NDMA/SDMA 24/7 toll-free lines (112, 1078, 1070) and verified disaster advisories.

6. Hackathon / Presentation Viva Questions & Ready Answers

Q1: What is the core innovation of ResQuick AI compared to Google Maps or Dial 112?

Answer: Commercial consumer maps only report slow traffic; they do not know that a 1.4-meter flood surge has cut off an ICU hospital approach, nor do they communicate with municipal departments. Dialing 112 creates isolated call logs where operators manually guess priorities. ResQuick AI dynamically aggregates multi-language citizen reports, applies an objective 5-factor mathematical risk model, ranks city sectors, and computes clear emergency ingress corridors to bypass blocked corridors.

Q2: How does the system handle citizens speaking in regional languages while officers need English?

Answer: When in distress, citizens must be able to speak their native mother tongue. ResQuick AI uses the HTML5 Web Speech Recognition API configured with regional dialect locales (such as kn-IN for Kannada or hi-IN for Hindi). The transcribed speech passes through a real-time semantic NLP translation engine that renders standardized administrative English in the Commander Admin Portal, eliminating linguistic barriers during crises.

Q3: How do you prevent fake, prank, or duplicate incident submissions?

Answer: ResQuick AI employs a three-tier defense: First, browser GPS bounding verifies that reports originate from real locations. Second, our spatial clustering engine aggregates complaints within a 500-meter radius into a single consolidated incident. Third, our client-side photo forensics check verifies natural lighting, structural fracture realism, and pixel continuity to eliminate AI-generated fakes.

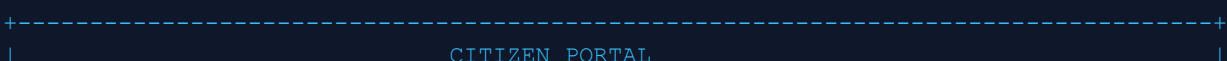
Q4: What happens if an area suddenly deteriorates, such as a culvert breach?

Answer: We demonstrate this live using the **Simulate Escalation Engine**. When cloudburst conditions worsen in Indiranagar, its score instantly jumps from 82 to 94+. The algorithm promotes Indiranagar to Priority #1, sounds an automated audio alarm, and executes automated resource reallocation—dynamically diverting trauma ambulances and heavy earthmovers from lower-risk depots directly to the hospital corridor.

Q5: How does the system function if internet connectivity is lost during a disaster?

Answer: ResQuick AI is built with an offline-first resilient architecture: map vectors and baseline perimeters are cached locally in browser storage, incident drafts are saved locally until a connection resumes, and an integrated SMS broadcast gateway allows commanders to dispatch mass emergency advisories over basic 2G/GSM cellular networks.

7. High-Level System Architecture Diagram



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| - Web Speech API (8 Regional Locales) - Reverse GPS Geocoding (Nominatim API) |
| - Real-time Language Preference Toggle - Photo & Video Evidence Upload |
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| REAL-TIME MULTILINGUAL NLP TRANSLATION ENGINE |
| (Kannada / Hindi / Tamil / Telugu / Bengali / Marathi → Administrative English) |
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| MATHEMATICAL RISK PRIORITIZATION & RESOURCE ALLOCATION CORE |
| Score = 0.30*Severity + 0.25*Pop + 0.20*Infra + 0.15*Access + 0.10*Facility |
| - Dynamic Re-sorting (#1 Priority) - Automated NDRF / Ambulance Reallocation |
| - Escalation Scenario Simulation - Explainable AI Rationales |
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| COMMANDER ADMIN OPERATIONS CENTER |
| - Standalone Full-Width Leaflet GIS Map (CartoDB Voyager & Satellite Layers) |
| - Smart Emergency Ingress Corridors - Live BBMP / NDMA Incident Table |
| - Formal Government Word Report Exporter - Emergency Siren Audio Dispatch Broadcast |
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